

Step 1: Calculate the Score & Prediction

$$\text{Score} = (0.5 \times 2) + (0.5 \times 1) = 1.5$$

$$\text{Prediction} = \text{sign}(1.5) = +1 \quad (\text{Prediction is correct})$$

Step 2: Calculate the Margin (How confident?)

$$\begin{aligned} \text{Margin} &= \text{Score} \times \text{Actual label} \\ &= 1.5 \times (+1) = 1.5 \end{aligned}$$

if margin of 1.5 $\rightarrow$ confidently correct

if margin is 0.2 (for example) $\rightarrow$ confidently wrong

Step 3: Calculate hinge loss

$$\text{loss} = \max(0, 1 - \text{Margin})$$

$$= \max(0, 1 - 1.5)$$

$$= \max(0, -0.5) = 0$$

Step 4: Calculate the Gradient

$\rightarrow$ if $\text{Margin} \geq 1$: Gradient = $[0, 0]$ $\rightarrow$ Do nothing
if $\text{Margin} < 1$: Gradient = $-(\text{Input}) \times (\text{Label})$.

Step 5: Update the weight

$$W_{\text{new}} = W_{\text{old}} - \eta \cdot \text{Gradient}$$

$$= [0.5, 0.5] - (0.1 \times [0, 0]) = [0.5, 0.5]$$